

Enhanced antibonding coupling enables optical transparency in p-type semiconductors by orbital engineering

Cite as: Appl. Phys. Lett. **129**, 012103 (2026); doi: [10.1063/5.0342323](https://doi.org/10.1063/5.0342323)

Submitted: 6 May 2026 · Accepted: 17 June 2026 ·

Published Online: 7 July 2026



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Export Citation



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ABSTRACT

Developing high-performance p-type transparent conducting materials (TCMs) remains challenging due to the intrinsic trade-off between a high valence band maximum (VBM), required for efficient acceptor ionization, and a wide bandgap for optical transparency. Motivated by the strongly p-type photovoltaic absorber $\text{Cu}_2\text{ZnSnS}_4$ (CZTS), we find that replacing Sn-5s with Si-3s orbital forming $\text{Cu}_2\text{ZnSiS}_4$ (CZSiS) significantly elevates the conduction band minimum through enhanced Si-3s/S-3p antibonding interaction, thereby enabling optical transparency. Meanwhile, CZSiS preserves the high VBM of CZTS, arising from Cu-3d/S-3p antibonding states, which facilitates the formation of Cu_{Zn}^- acceptors, analogous to CZTS. First-principles calculations demonstrate that CZSiS exhibits intrinsic p-type transparent conductivity with an electrical conductivity of 6.53 S/cm along with an optical transmittance of $\sim 80\%$. The isostructural compound $\text{Cu}_2\text{MgSiS}_4$ (CMSiS) further confirms the generality of this approach. This study proposes CZSiS and CMSiS as promising candidates for high-performance p-type TCMs and highlights orbital engineering as an effective strategy for transforming p-type semiconductors into efficient TCMs.

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Transparent conducting materials (TCMs), essential for transparent electronics,¹ combine high conductivity with visible-light transparency. Currently, n-type TCMs have achieved this balance and dominate commercial applications,^{2–4} while p-type TCMs face a fundamental challenge that a wide bandgap required for optical transparency typically leads to a low valence band maximum (VBM), suppressing acceptor ionization and resulting in a critically low hole concentration. This trade-off has made high-performance p-type TCMs exceptionally rare, posing a persistent bottleneck in optoelectronics.

Significant efforts have been devoted to developing high-performance p-type TCMs. Among these, CuI has emerged as a promising candidate, exhibiting excellent p-type conductivity enabled by its high-energy and delocalized VBM arising from Cu- t_{2g} /I-5p antibonding states.⁵ However, the antibonding character of Cu-I bond promotes rapid Cu migration, leading to poor environmental stability against air and moisture.⁶ Conventional oxides generally

exhibit robust chemical stability, but their low VBMs hinder acceptor ionization, limiting the realization of p-type TCMs. A distinct class of CuO-based compounds, represented by delafossites $\text{CuM}_{\text{III}}\text{O}_2$ ($M_{\text{III}} = \text{Al, Ga, In, Sc}$) and SrCu_2O_2 , exploit filled Cu-3d¹⁰ orbitals coupled with O-2p states to achieve a high VBM while maintaining a relatively wideband gap.^{7,8} Similarly, octahedrally coordinated d⁶ transition metals in spinel $\text{Zn}(\text{M}_{\text{VIII}})_2\text{O}_4$ ($M_{\text{VIII}} = \text{Co}^{3+}, \text{Rh}^{3+}, \text{Ir}^{3+}$)⁹ exhibit crystal field splitting into fully occupied t_{2g} and empty e_g states, which predominantly form the VBM and conduction band minimum (CBM), respectively. To mitigate valence band localization arising from O-2p states, bi-anion quaternary oxychalcogenides LnCuOCh ($\text{Ln} = \text{La, Y, Pr, Nd}$ and $\text{Ch} = \text{S, Se, Te}$) have been designed.¹⁰ These compounds feature alternating $[\text{Ln}_2\text{O}_2]^{2+}$ and $[\text{Cu}_2\text{Ch}_2]^{2-}$ layers, in which Cu-Ch antibonding states dominate the VBM. Recently, the pure sulfide GaAgS_2 has been proposed as a p-type TCM with markedly enhanced hole conductivity. However, attaining a bandgap that exceeds the visible transparency range

necessitates external strain, thereby limiting its practical applicability.¹¹ Compounds such as BaGa_2Se_4 and CdAl_2Te_4 , enabled by orbital coupling and anion synergy,¹² as well as alkali-doped wurtzite GaN ,¹³ have also been predicted as promising p-type TCM candidates. Additionally, several optically transparent binary compounds, including phosphides,¹⁴ sulfides,¹⁵ selenides,¹⁶ and halides¹⁷ possess high VBMs, yet they are rarely reported as p-type TCMs in practice due to low hole concentrations or structural instability. High-throughput screening has identified CaTe and Li_3Sb as promising candidates,¹⁸ while their indirect bandgaps are below the optical transparency threshold. Overall, the development of p-type TCMs remains constrained by the intrinsic trade-offs among electrical conductivity and optical transparency. Recently, we note that kesterite $\text{Cu}_2\text{ZnSnS}_4$ (CZTS),¹⁹ a quaternary sulfide widely used as a photovoltaic absorber, exhibits strong p-type conductivity. Its elevated VBM, arising from Cu-3d/S-3p antibonding interactions, facilitates acceptor ionization in accordance with the doping limit rule,²⁰ while the CBM is primarily derived from low-energy Sn-5s/S-3p antibonding states, leading to a narrow bandgap of 1.50 eV. Accordingly, we speculate that substituting Sn-5s with an isoelectronic element possessing stronger antibonding interactions with S-3p orbital could widen the bandgap into the optical transparency range while preserving the favorable high VBM as in CZTS, thereby enabling effective p-type transparent conductivity.

In this work, we find that replacing Sn with Si in CZTS yields transparent $\text{Cu}_2\text{ZnSiS}_4$ (CZSiS) while preserving its pronounced p-type conductivity. The bandgap increases from 1.50 to 3.04 eV by raised CBM, attributable to the strong Si-3s/S-3p antibonding coupling. The intrinsically strong p-type conductivity, similar to that of kesterite CZTS, originates from the high VBM induced by Cu-3d/S-3p antibonding interactions. This promotes the ionization of low-energy Cu_{Zn}^- acceptors, yielding a high hole concentration exceeding 10^{18} cm^{-3} , an electrical conductivity of 6.53 S/cm, and a visible-light transmittance around 80% at a thickness of 100 nm. The isostructural compound $\text{Cu}_2\text{MgSiS}_4$ (CMSiS) also demonstrates promising performance, achieving a conductivity of 1.14 S/cm and a transmittance above 80%.

The total energies and electronic structures of the pristine and defective systems are calculated based on a $2 \times 2 \times 2$ supercell containing 128 atoms. First-principles calculations are performed with the Vienna *ab initio* simulation package (VASP),^{21,22} employing the standard frozen-core project-augmented-wave (PAW) method²³ with the Heyd–Scuseria–Ernzerhof (HSE06) functional.²⁴ The mixing parameters of 0.34, 0.39, 0.34, and 0.29 are adopted to reproduce the experimental bandgap of CZTS (1.50 eV),²⁵ $\text{Cu}_2\text{ZnGeS}_4$ (CZGS: 2.25 eV),²⁵ CZSiS (3.04 eV),²⁶ and CMSiS (3.20 eV).²⁷ All atomic structures are fully relaxed until the Hellmann–Feynman forces on each atom are below 0.01 eV/Å and the total energy is converged to 1×10^{-5} eV. A plane wave cutoff energy of 450 eV and a Γ -centered $2 \times 2 \times 2$ k-point mesh are used. The optimized lattice parameters of CZSiS ($a = 7.38$ Å, $b = 6.46$ Å, and $c = 6.16$ Å) and CMSiS ($a = 7.59$ Å, $b = 6.45$ Å, and $c = 6.21$ Å) are in good agreement with experimental values (CZSiS: $a = 7.44$ Å, $b = 6.40$ Å, and $c = 6.14$ Å and CMSiS: $a = 7.56$ Å, $b = 6.44$ Å, and $c = 6.18$ Å).^{26,27} The ShakeNBreak method²⁸ is employed to generate initial defective structures. Finite-size effects for charged defects are corrected using the Kumagai–Oba correction, including potential alignment, as implemented in

Doped.²⁹ The convergence of defect formation energies is examined using a larger $3 \times 2 \times 2$ supercell and a denser $3 \times 3 \times 3$ k-point mesh, respectively, for the dominant Cu_{Zn}^- and Cu_{Mg}^- acceptors, yielding results consistent with those adopted in this work (see part I of the [supplementary material](#)). Band alignment is calculated using the three-step method.³⁰ The hole electrical conductivity and mobility are calculated using the *ab initio* scattering and transport (AMSET).³¹ The phonon dispersions are calculated using the density functional perturbation theory method combined with first-principles calculations, as implemented in the Phonopy package.³²

The strongly p-type photovoltaic absorber CZTS is typically described in the $I\bar{4}$ space group. Its electronic structure and total energy are very close to those of the $\text{Pmn}2_1$ phase due to their equivalent chemical coordination environments (see part II of the [supplementary material](#)). For consistency with $\text{Pmn}2_1$ structured CZGS, CZSiS, and CMSiS, the compared CZTS is modeled in the $\text{Pmn}2_1$ structure, as shown in Fig. 1(a).

Band alignment analysis provides an intuitive preliminary assessment of p-type conductivity based on the doping limit rule.²⁰ Using the three-step method,³⁰ the VBM positions of CZGS, CZSiS, and CMSiS are found to decrease sequentially while remaining comparable to that of CZTS, as shown in Fig. 1(b) (see part III of the [supplementary material](#) for calculating details). This means that substituting Sn with Si or Ge could also facilitate acceptor ionization and preserve the strong p-type characteristics as in CZTS. In contrast, the CBM is significantly upshifted in the substituted structures, yielding bandgaps of 2.25, 3.04, and 3.20 eV for CZGS, CZSiS, and CMSiS, respectively, as illustrated by the blue histograms of Fig. 1(b).

In all four systems, the CBM and VBM originate from antibonding interactions of $X\text{-ns}/\text{S-3p}$ ($X\text{-ns}$: Sn-5s, Ge-4s, and Si-3s) and Cu-3d/S-3p states,³³ respectively. Shorter bond lengths enhance antibonding coupling, thereby raising the corresponding antibonding states. To elucidate the CBM upshift, bond length analysis shows that the Sn–S, Ge–S, and Si–S bonds decrease progressively, consistent with the observed CBM increase and consequent bandgap enlargement [Fig. 1(c)]. In contrast, the nearly invariant Cu–S bond lengths across these systems account for the minimal variation in VBM positions.

To further isolate the lattice effect, we replaced the Ge/Si with Sn while fixing the lattice constants of CZGS, CZSiS, and CMSiS. The resulting CBM positions remain comparable to those of the Ge- and Si-based systems, as shown by the green histograms in Fig. 1(b). This indicates that the reduced lattice constants, relative to CZTS, compress the Ge–S and Si–S bond lengths (see part IV of the [supplementary material](#)), thereby strengthening the Ge-4s/S-3p and Si-3s/S-3p antibonding interactions. Here, the standard HSE06 functional with a fixed mixing parameter of 0.25 is adopted for all four compounds, confirming that the obtained conclusions are robust and independent of the optimized mixing parameter (see part V of the [supplementary material](#)).

Figures 2(a)–2(d) presents the orbital-projected band structures of X ($X = \text{Sn, Ge, and Si}$) and the corresponding projected crystal orbital Hamilton populations (-pCOHP) for CZTS, CZGS, CZSiS, and CMSiS. More negative -pCOHP values indicate stronger antibonding interactions. The CBM is dominated by antibonding coupling between $X\text{-s}$ and S-3p orbitals. With progressive substitution of Sn by Ge and Si, this antibonding interaction is strengthened, as evidenced by increasingly negative -pCOHP values near the CBM,

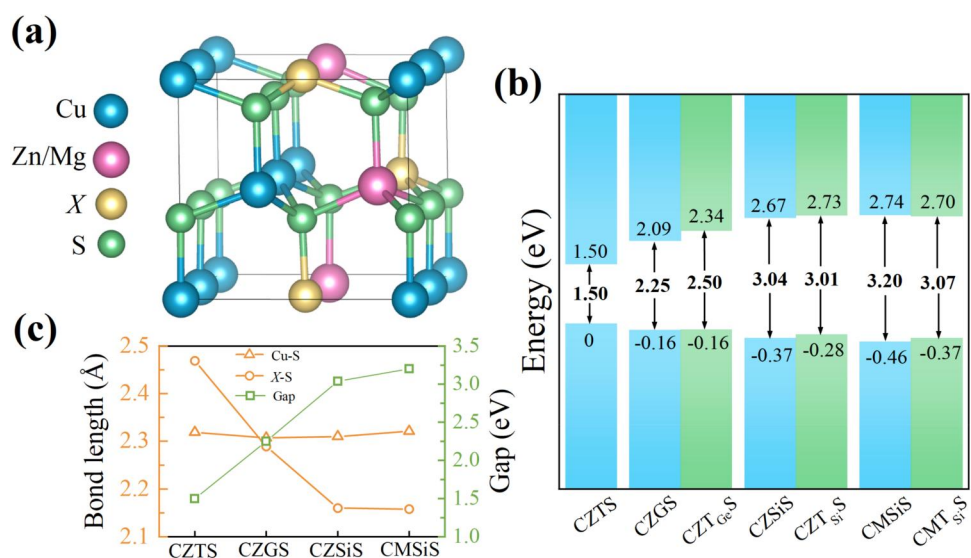


FIG. 1. Atomic structures, band alignment, bond lengths, and bandgaps. (a) Atomic structures of CZXS and CMXS in the Pmn21 space group (X = Sn, Ge, and Si). (b) Band alignment among CZTS, CZGS, CZSiS, and CMSiS. Their experimental bandgaps are collected from Refs. 25–27. (c) Bond lengths of Cu–S and X–S, and bandgaps of the four systems. CZT_XS denotes the CZXS structure with Sn substitution at the X site.

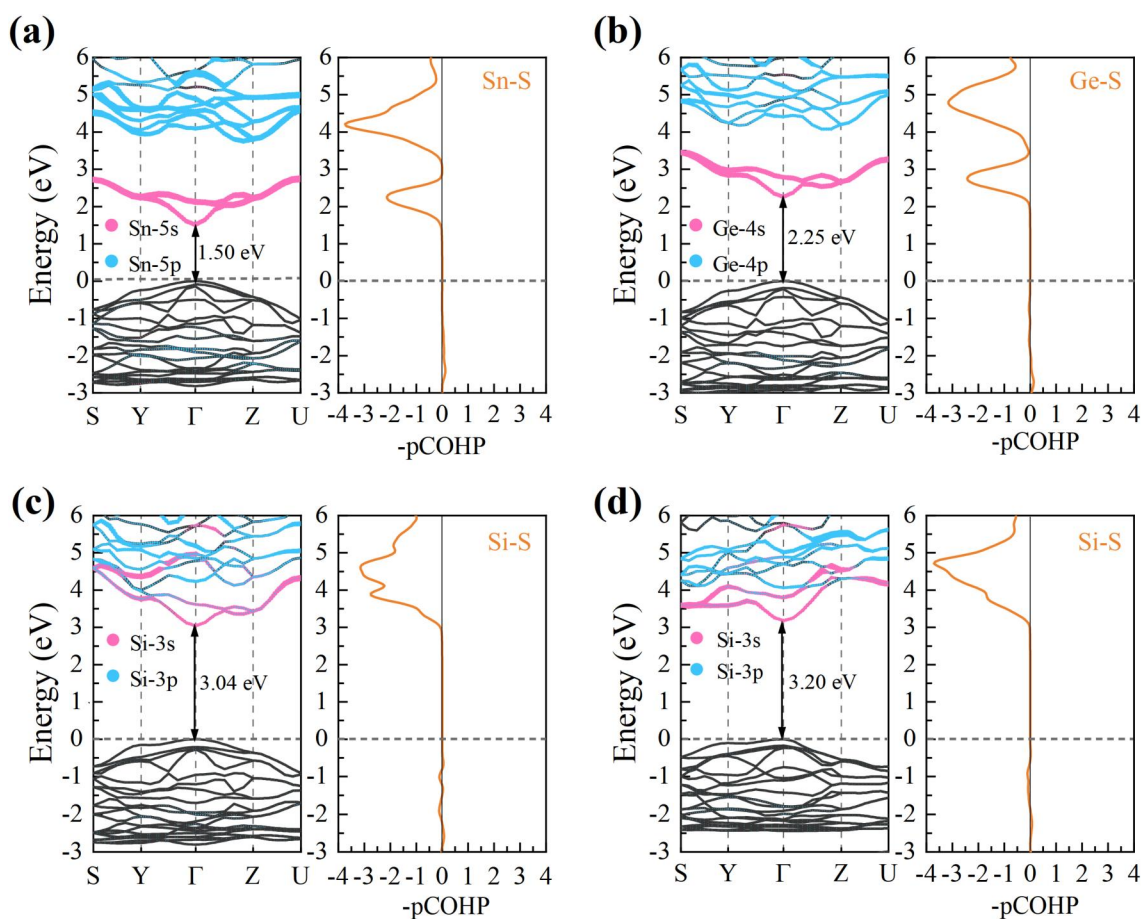


FIG. 2. Orbital-projected band structures and projected crystal orbital Hamilton populations (pCOHP). (a)–(d) Orbital projections of Sn, Ge, Si, and Si in CZTS, CZGS, CZSiS, and CMSiS, respectively, along with their antibonding contributions with S as reflected by -pCOHP.

resulting in its upward shift. This enhanced antibonding interaction originates from the increased orbital overlap between X-s and S-3p states, as confirmed by the progressively larger antibonding crystal orbital overlap population (COOP) from CZTS to CMSiS. Consistently, the partial charge density of the CBM exhibits reduced electron states on the X-S bonds, indicating stronger antibonding character and a larger bonding-antibonding splitting (see part VI of the [supplementary material](#)). Consequently, the bandgaps of CZSiS and CMSiS reach 3.04 and 3.20 eV, respectively, placing them near or within the visible transparency range (3.10 eV). These two systems are therefore selected for further investigation of p-type conductivity and optical transparency.

The conduction type of semiconductors is mainly governed by the competition between energetically favorable ionized donor and acceptor defects, whose formation energies depend on both the Fermi level³⁴ and elemental chemical potentials.³⁵ Generally, the Fermi level resides between the VBM and CBM. Under heavy doping conditions, it may shift close to VBM or CBM for p-type or n-type conductivity, respectively. The allowable range of elemental chemical potentials is constrained by the stable phase diagram as shown in [Figs. 3\(a\)](#) and [3\(d\)](#) (see part VII of the [supplementary material](#) for details on phase diagram calculations). Phonon dispersion calculations and *ab initio*

molecular dynamics (AIMD) simulations further confirm the dynamical and thermal stability of the CZSiS and CMSiS compounds (see part VIII of the [supplementary material](#)).

The formation energy, $\Delta H(\alpha, q)$, of a defect α in charge state q is defined as^{36,37}

$$\Delta H(\alpha, q) = E(\alpha, q) - E(\text{host}) + \sum_i n_i (\mu_i^0 + \mu_i) + q(E_{\text{VBM}} + E_F) + \Delta_{\text{ES}},$$

where $E(\alpha, q)$ represents the total energy of defect α in charge state q within a $2 \times 2 \times 2$ supercell model, and $E(\text{host})$ corresponds to the total energy of an identical supercell without defects. The Fermi level (E_F) is referenced to the host material's VBM (E_{VBM}). The parameter n_i describes the atomic imbalance of element i between the defective and pristine systems, where positive values indicates atom removal and negative values corresponds to the addition of $|n_i|$ atoms. The correction term Δ_{ES} , computed using the Doped software,²⁹ compensates for spurious electrostatic interactions between periodic images, thereby improving the accuracy of charged-defect formation energies.

Point C (Cu-moderate and Zn-poor), which most favors the formation of the acceptor defect Cu_{Zn}^- , and point M (Cu-rich,

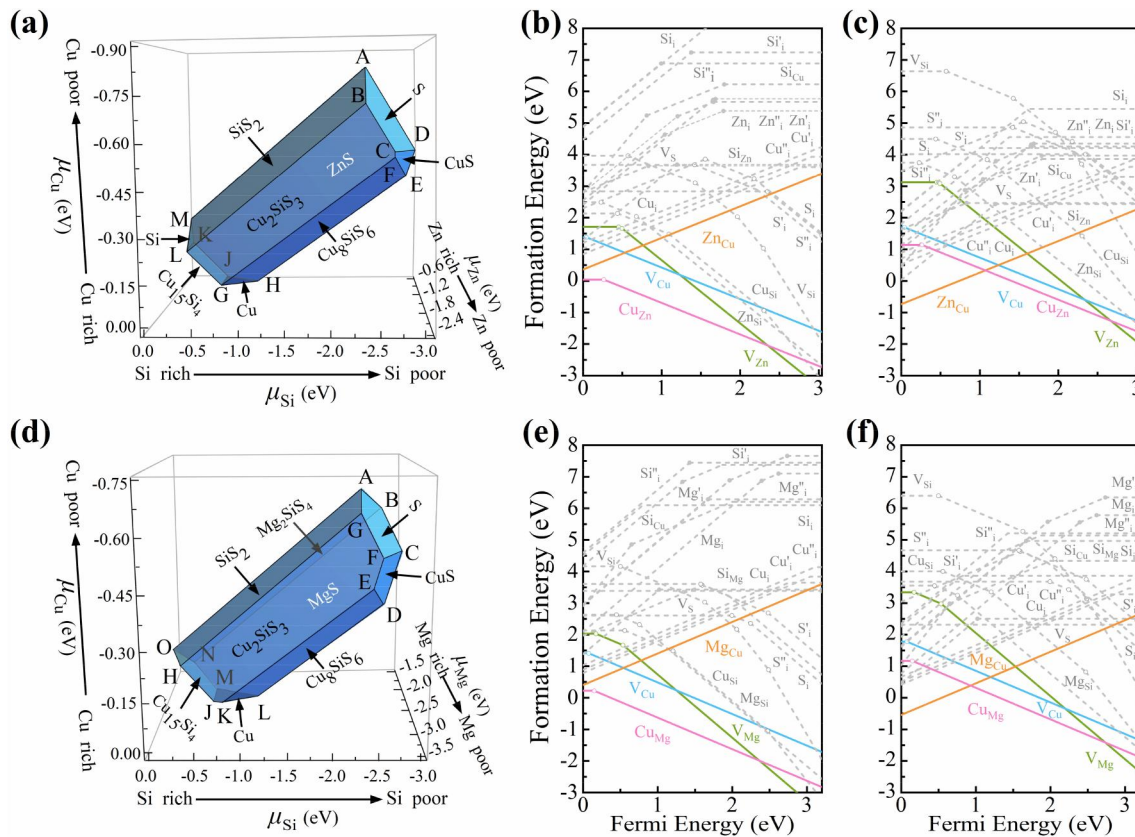


FIG. 3. Phase diagrams and defect formation energies. (a) and (d) Stable phase diagram showing the allowable ranges of elemental chemical potentials for CZSiS and CMSiS, respectively. (b) and (c) Formation energies of intrinsic defects as a function of Fermi level under point C (Cu-moderate, Zn-poor) and point M (Cu-rich, Zn-rich) conditions, as indicated in (a). (e) and (f) Formation energies of intrinsic defects as a function of Fermi level under point E (Cu-moderate, Mg-poor) and point N (Cu-rich, Mg-rich) conditions, as indicated in (d). Solid lines denote low-energy defects, and dashed lines denote high-energy defects.

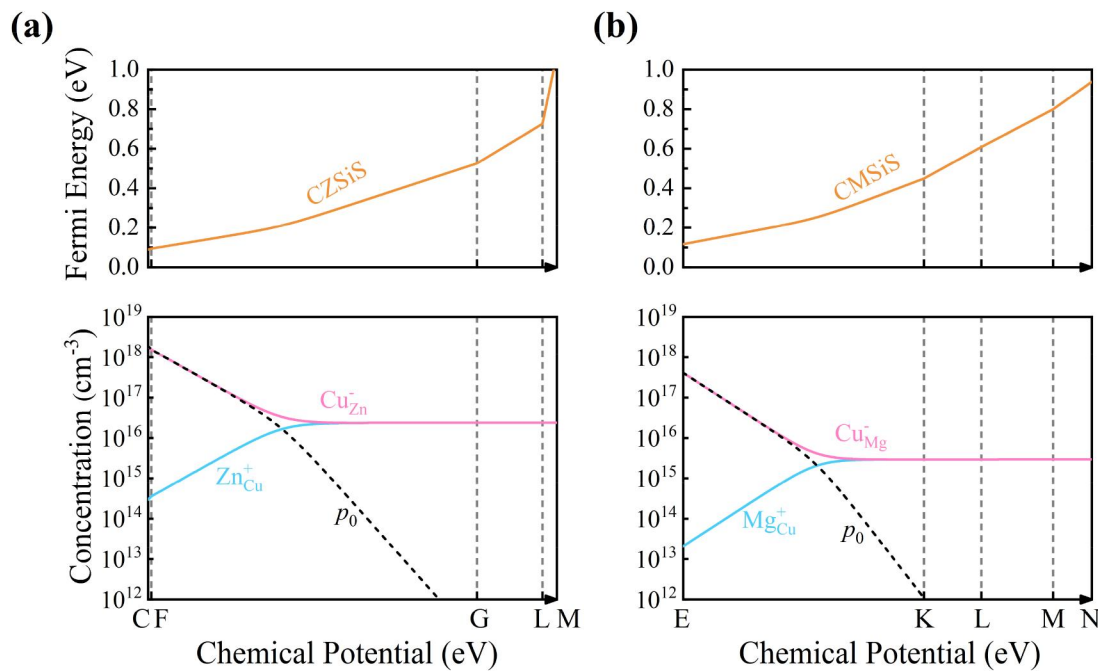


FIG. 4. Fermi level, defect and hole concentrations. Calculated Fermi level (upper panels), and defect/hole concentrations (lower panels) as functions of the elemental chemical potential at ambient temperature for (a) CZSiS and (b) CMSiS. The chemical potential range is taken along the extreme points of the phase stability region, following the paths C–F–G–L–M for CZSiS and E–K–L–M–N for CMSiS.

Zn-rich), where its formation is least favorable, are identified in Fig. 3(a) and selected as representative chemical potential conditions for defect formation energy calculations of CZSiS, as shown in Figs. 3(b) and 3(c), respectively. At point C, the Cu_{Zn}^- acceptor exhibits an exceptionally low formation energy with a shallow transition level (0.26 eV above VBM) and overwhelmingly dominates over donors (see part IX of the [supplementary material](#) for the transition energy levels of all defects). Even at point M, the Fermi level is determined by the intersection of the formation energies of Cu_{Zn}^- and Zn_{Cu}^+ , and remains pinned within the p-type region. For CMSiS, points E (Cu-moderate and Mg-poor) and N (Cu-rich and Mg-rich) conditions, shown in Fig. 3(d), are selected for defect formation energy calculations, as shown in Figs. 3(e) and 3(f). Similar to the Cu_{Zn}^- defect in CZSiS, the Cu_{Mg}^- acceptor dominates the p-type conductivity in this system with a shallow transition level (0.15 eV above VBM), although its formation energy is slightly higher than that of Cu_{Zn}^- in CZSiS. This difference arises from the relatively larger ionic radius mismatch between Cu^+ (0.60 Å) and Mg^{2+} (0.57 Å), compared with that between Cu^+ and Zn^{2+} (0.60 Å),³⁸ as well as the slightly higher VBM in CZSiS as shown in Fig. 1(b). Defect formation energies under the remaining chemical potential limiting points in the phase diagrams shown in Figs. 3(a) and 3(d) and more clear defect formation energies in Figs. 3(b), 3(c), 3(e), and 3(f) are provided in part X of the [supplementary material](#).

The position of Fermi level relative to the mid-gap provides an intuition of the conductive type in semiconductors. Under thermodynamic equilibrium, the Fermi level can be determined by self-consistently solving the charge neutrality equation at a given temperature.³⁹ Substituting the obtained Fermi level into the expressions

for defect and carrier concentrations then yields the corresponding defect and carrier concentrations.^{38,40} (see part XI of the [supplementary material](#) for detailed calculations of effective mass, Fermi level, charge neutrality equation, and defect and hole concentrations).

Figure 4(a) presents the Fermi level and hole concentration in CZSiS, determined by the dominant defects under chemical synthesis conditions along the C–F–G–L–M line at 300 K. The Fermi level reaches its minimum at point C (Cu-moderate, Zn-poor), where the concentration of Cu_{Zn}^- defects and the hole concentration attain values as high as $1.88 \times 10^{18} \text{ cm}^{-3}$. For CMSiS, as shown in Fig. 4(b), the lowest Fermi level occurs at point E (Cu-moderate, Mg-poor), with the corresponding Cu_{Mg}^- defect and hole concentrations reaching $4.11 \times 10^{17} \text{ cm}^{-3}$. These results indicate that the Zn-poor and Mg-poor while Cu-moderate conditions in CZSiS and CMSiS promote the formation of Cu_{Zn}^- and Cu_{Mg}^- acceptors, respectively, leading to appreciable hole concentrations, similar to the promoted formation of Cu_{Zn}^- defects in CZTS under Zn-poor condition.⁴¹

The hole electrical conductivity and mobility of CZSiS and CMSiS are calculated using the *ab initio* scattering and transport (AMSET) method.³¹ As shown in Figs. 5(a) and 5(d), the maximum electrical conductivities of CZSiS and CMSiS are 6.53 and 1.14 S/cm, respectively, under their highest hole concentrations. Analysis of their mobilities, as shown in Figs. 5(b) and 5(e), indicates that charge transport in both systems is primarily limited by polar optical phonon (POP) scattering, whereas acoustic deformation potential (ADP) and ionized impurity (IMP) scattering have only minor effects. The optical transmittance (T) is calculated using $T(E, t) = (1 - R)e^{-\alpha(E)t}$,⁴² where R is the reflectivity, α is the absorption coefficient as a function of incident photon energy E , and t is the material thickness. At a thickness

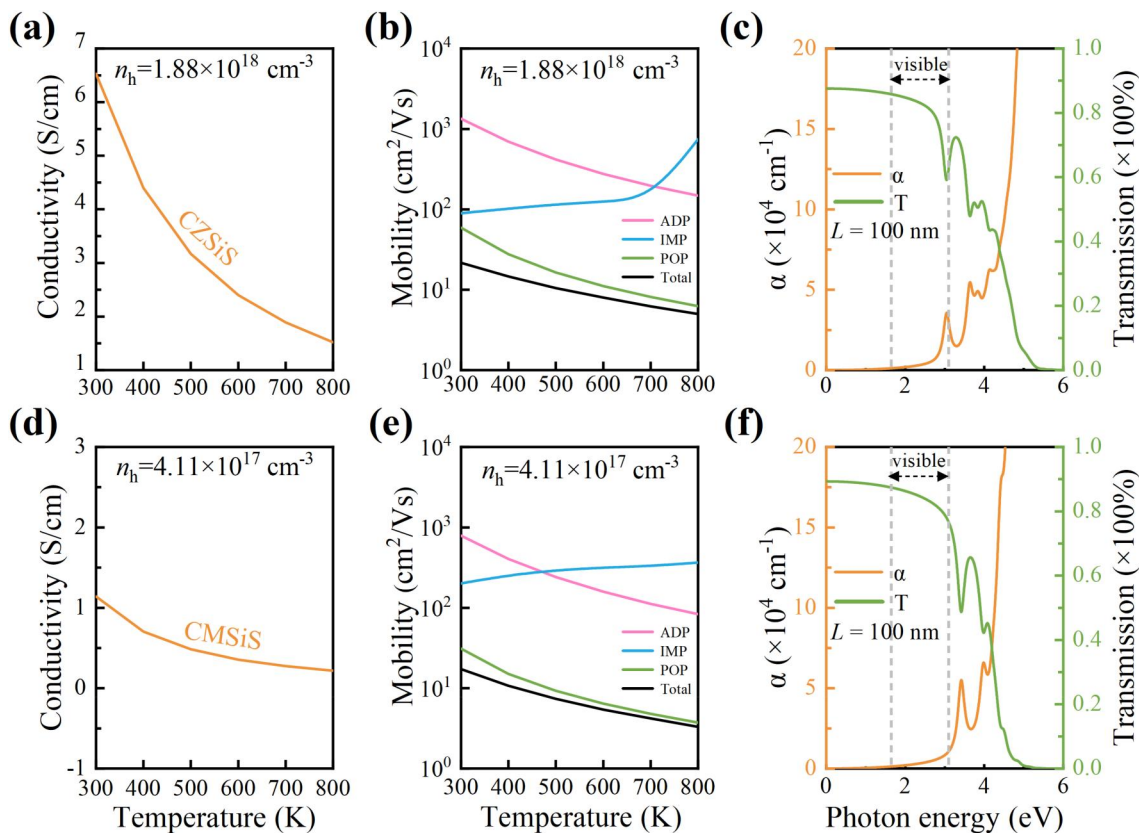


FIG. 5. Electrical conductivity and optical transparency. (a) and (d) Temperature-dependent electrical conductivities of CZSiS and CMSiS, respectively, at their highest hole concentrations. (b) and (e) Hole mobilities of CZSiS and CMSiS, respectively, considering scattering from acoustic deformation potential (ADP), ionized impurity (IMP), and polar optical phonon (POP). (c) and (f) Optical absorption coefficients and transmittance of CZSiS and CMSiS, respectively, at a thickness of 100 nm.

of 100 nm, CZSiS exhibits transmittance above 80% across most of the visible spectrum, decreasing to approximating 60% only within a narrow region near the ultraviolet, as shown in Fig. 5(c). This behavior is attributed to its bandgap being slightly smaller than the 3.10 eV required for full visible-light transparency. Under the same thickness, CMSiS maintains transmittance above 80% throughout the visible region, as shown in Fig. 5(f). Additionally, both systems show low

absorption coefficients in the visible-light range. Table I compares the materials investigated in this work with commonly experimentally studied p-type TCMs (e.g., CuAlO₂, ZnIr₂O₄, NiO, LaCuOS, SrCu₂O₂, and CuI) under conditions without external doping or annealing. Table SIV further summarizes the theoretical transport properties of these reference materials, providing a more consistent theoretical benchmark for evaluating the intrinsic transport

TABLE I. A comparison of optical transmittance (T), bandgap (E_g), electrical conductivity (σ), carrier concentration (n), and mobility (μ) between experimentally synthesized p-type TCMs of various thickness (d) and the theoretically predicated CZSiS and CMSiS in this work.

Material	Deposition	d (nm)	T (%)	E_g (eV)	σ (S/cm)	n (cm ⁻³)	μ (cm ² /Vs)	Ref.
CZSiS	...	100	80	3.04	6.53	1.88×10^{18}	21.6	This work
CMSiS	...	100	80	3.20	1.14	4.11×10^{17}	17.3	This work
CuAlO ₂	PLD	500	80	3.50	1	1.3×10^{17}	10.4	44
ZnIr ₂ O ₄	PLD	100	60	2.97	2.09	...	...	45
NiO	Mag. sputtering	110	40	3.80	7.1	1.3×10^{19}	...	46
LaCuOS	RFMS	800	55	3.1	0.115	6.55×10^{18}	0.11	47
SrCu ₂ O ₂	PLD	220–300	80	3.3	0.0065	1.0×10^{13}	...	48
CuI	MBE	283	...	3.1	0.89	1.06×10^{18}	110	49

performance of CZSiS and CMSiS (see part XII of the [supplementary material](#)). It can be seen that, solely through intrinsic defects tuned by elemental chemical potentials, CZSiS and CMSiS exhibit pronounced advantages in simultaneously achieving high optical transparency and electrical conductivity relative to those well-known p-type TCMs. Additionally, CZSiS and CMSiS have been experimentally synthesized via sulfurization annealing and conventional high-temperature solid-state reactions, respectively.^{26,27,43} Combined with the finite thermodynamic stability regions predicted in this work, these results suggest that both compounds are experimentally realizable without a significant risk of phase separation under suitable growth conditions.

This work employs an orbital engineering strategy to provide a viable pathway for resolving the long-standing trade-off between achieving a high valence band maximum (VBM) for efficient hole transport and maintaining a wide bandgap for optical transparency. Inspired by the strong p-type photovoltaic absorber CZTS, we demonstrate this concept in CZSiS, obtained by substituting Sn-5s orbitals in CZTS with Si-3s orbitals. This substitution significantly enhances antibonding coupling between cation s and anion p states, elevating the CBM to widen the bandgap into the transparent regime, and preserves a high VBM through strong Cu-S antibonding interactions, facilitating acceptor ionization, analogous to CZTS. As a result, CZSiS exhibits intrinsic p-type conductivity (6.53 S/cm) along with an optical transmittance of ~80%. The isostructural compound CMSiS further confirms the generality of this approach. Overall, this study identifies CZSiS and CMSiS as promising candidates for high-performance p-type transparent conducting materials (TCMs) and highlights orbital engineering as an effective strategy for transforming intrinsically p-type semiconductors into efficient TCMs.

See the [supplementary material](#) for detailed descriptions of the convergence tests for supercell size and k-point sampling, and calculations of band structure, band alignment, bond length statistics, phase diagrams, defect formation energies, phase diagrams, defect formation energy, effective mass, Fermi level, charge neutrality equation, and defect and hole concentrations.

This work was sponsored by the Natural Science Foundation of Henan; the Henan Province Outstanding Youth Science Foundation (Grant No. 252300421215); Key Scientific Research Project of Higher Education Institutions in Henan Province—Basic Research Program (Grant No. 25ZX005); the National Natural Science Foundation of China (Grant No. 12274114); and the 2026 Annual Project of College Students' Innovation and Entrepreneurship Training Program (Grant No. DCZ2026034).

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Xiaobei Lei: Methodology (equal); Writing – original draft (equal). **Jinping Zhang:** Methodology (equal); Writing – original draft (equal). **Wentao Yang:** Conceptualization (equal); Investigation (equal). **Ke Zhao:** Conceptualization (equal); Methodology (equal); Writing – original draft (equal). **Xinchun Li:** Conceptualization

(equal); Investigation (equal). **Chengyan Liu:** Funding acquisition (equal); Investigation (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

REFERENCES

- E. Fortunato, P. Barquinha, and R. Martins, "Oxide semiconductor thin-film transistors: A review of recent advances," *Adv. Mater.* **24**(22), 2945–2986 (2012).
- H. Ohta, M. Orita, M. Hirano, H. Tanji, H. Kawazoe, and H. Hosono, "Highly electrically conductive indium-tin-oxide thin films epitaxially grown on yttria-stabilized zirconia (100) by pulsed-laser deposition," *Appl. Phys. Lett.* **76**(19), 2740–2742 (2000).
- M. Izaki and T. Omi, "Transparent zinc oxide films prepared by electrochemical reaction," *Appl. Phys. Lett.* **68**(17), 2439–2440 (1996).
- C. Agashe, J. Hüpkens, G. Schöpe, and M. Berginski, "Physical properties of highly oriented spray-deposited fluorine-doped tin dioxide films as transparent conductor," *Sol. Energy Mater. Sol. Cells* **93**(8), 1256–1262 (2009).
- J. Willis, R. Claes, Q. Zhou, M. Giantomassi, G.-M. Rignanese, G. Hautier, and D. O. Scanlon, "Limits to hole mobility and doping in copper iodide," *Chem. Mater.* **35**(21), 8995–9006 (2023).
- A. N. Fioretti and M. Morales-Masis, "Bridging the p-type transparent conductive materials gap: Synthesis approaches for disperse valence band materials," *J. Photonics Energy* **10**(4), 042002 (2020).
- T. Arnold, D. J. Payne, A. Bourlange, J. P. Hu, R. G. Egdell, L. F. J. Piper, L. Colakerol, A. De Masi, P.-A. Glans, T. Learmonth, K. E. Smith, J. Guo, D. O. Scanlon, A. Walsh, B. J. Morgan, and G. W. Watson, "X-ray spectroscopic study of the electronic structure of CuCrO₂," *Phys. Rev. B* **79**(7), 075102 (2009).
- W. Yang, J. Zhang, K. Zhao, X. Li, X. Lei, and C. Liu, "Electronic origins of p-type transparent conductivity in SrCu₂O₂," *Appl. Phys. Lett.* **128**(16), 161905 (2026).
- M. N. Amini, H. Dixit, R. Saniz, D. Lamoén, and B. Partoens, "The origin of p-type conductivity in ZnM₂O₄ (M = Co, Rh, Ir) spinels," *Phys. Chem. Chem. Phys.* **16**(6), 2588 (2014).
- H. Hiramatsu, K. Ueda, K. Takafuji, H. Ohta, M. Hirano, T. Kamiya, and H. Hosono, "Intrinsic excitonic photoluminescence and band-gap engineering of wide-gap p-type oxychalcogenide epitaxial films of LnCuOCh (Ln = La, Pr, and Nd; Ch = S or Se) semiconductor alloys," *J. Appl. Phys.* **94**(9), 5805–5808 (2003).
- Y. Deng, W. Chen, L. Yang, G. Lu, and S. Fan, "Strain greatly improves the visible light transparency and p-type electrical conductivity for GaAgS₂," *Ceram. Int.* **52**, 26888 (2026).
- J.-Y. Xia, D.-H. Fan, Z.-T. Liu, and Q.-J. Liu, "Orbital coupling–anion synergy in AB₂Ch₄ semiconductors: Band engineering for high-performance p-type transparent conductive materials," *J. Mater. Chem. C* **14**(11), 4330–4336 (2026).
- J.-Y. Xia, W. Zeng, Z.-T. Liu, J. Gao, and Q.-J. Liu, "P-type conductivity mechanism and element doping modification in wurtzite GaN from first-principles calculations," *Cryst. Growth Des.* **25**(14), 5299–5311 (2025).
- A. Crovetto, J. M. Adamczyk, R. R. Schnepf, C. L. Perkins, H. Hempel, S. R. Bauers, E. S. Toberer, A. C. Tamboli, T. Unold, and A. Zakutayev, "Boron phosphide films by reactive sputtering: Searching for a p-type transparent conductor," *Adv. Mater. Inter.* **9**, 2200031 (2022).
- Y. Chen, S. W. Fan, and G. Y. Gao, "Theoretical insights into the defect performance of the wide bandgap semiconductor BaS," *Phys. Chem. Chem. Phys.* **25**(16), 11745–11755 (2023).
- S. W. Fan, Y. Chen, and L. Yang, "Transparency and p-type conductivity of BeSe doped with group VA atoms: A hybrid functional study," *J. Phys. Chem. C* **126**, 19446–19454 (2022).

- ¹⁷H. Yu, X. Cai, Y. Yang, Z.-H. Wang, and S.-H. Wei, "Band gap anomaly in cuprous halides," *Comput. Mater. Sci.* **203**, 111157 (2022).
- ¹⁸V.-A. Ha, G. Yu, F. Ricci, D. Dahliah, M. J. Van Setten, M. Giantomassi, G.-M. Rignanese, and G. Hautier, "Computationally driven high-throughput identification of CaTe and Li₃Sb as promising candidates for high-mobility p-type transparent conducting materials," *Phys. Rev. Mater.* **3**(3), 034601 (2019).
- ¹⁹H. Xiang, Z. Zheng, K. Zhao, C. Liu, and J. Zhao, "Interfacial passivation of kesterite solar cells for enhanced carrier lifetime: Ab initio nonadiabatic molecular dynamics study," *Adv. Funct. Mater.* **34**(46), 2407991 (2024).
- ²⁰Y. Yan and S. Wei, "Doping asymmetry in wide-bandgap semiconductors: Origins and solutions," *Phys. Status Solidi B* **245**(4), 641–652 (2008).
- ²¹G. Kresse and J. Furthmüller, "Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set," *Comput. Mater. Sci.* **6**(1), 15–50 (1996).
- ²²G. Kresse and J. Furthmüller, "Efficient iterative schemes for *ab initio* total-energy calculations using a plane-wave basis set," *Phys. Rev. B* **54**(16), 11169–11186 (1996).
- ²³G. Kresse and D. Joubert, "From ultrasoft pseudopotentials to the projector augmented-wave method," *Phys. Rev. B* **59**(3), 1758–1775 (1999).
- ²⁴A. V. Krukau, O. A. Vydrov, A. F. Izmaylov, and G. E. Scuseria, "Influence of the exchange screening parameter on the performance of screened hybrid functionals," *J. Chem. Phys.* **125**(22), 224106 (2006).
- ²⁵K. Tsuji, T. Maeda, and T. Wada, "Optical properties and electronic structures of Cu₂ZnSnS₄, Cu₂ZnGeS₄, and Cu₂Zn(Ge,Sn)S₄ and Cu₂Zn(Ge,Sn)Se₄ solid solutions," *Jpn. J. Appl. Phys. Part 1* **57**(8S3), 08RC21 (2018).
- ²⁶G. Q. Yao, H. S. Shen, E. D. Honig, R. Kershaw, K. Dwight, and A. Wold, "Preparation and characterization of the quaternary chalcogenides Cu₂B(II)C(IV)X₄[B(II) = Zn, Cd; C(IV) = Si, Ge; X = S, Se]," *Solid State Ionics* **24**, 249–252 (1987).
- ²⁷B.-W. Liu, M.-J. Zhang, Z.-Y. Zhao, H.-Y. Zeng, F.-K. Zheng, G.-C. Guo, and J.-S. Huang, "Synthesis, structure, and optical properties of the quaternary diamond-like compounds I₂–II–IV–VI₄ (I = Cu; II = Mg; IV = Si, Ge; VI = S, Se)," *J. Solid State Chem.* **204**, 251–256 (2013).
- ²⁸I. Mosquera-Lois, S. R. Kavanagh, A. Walsh, and D. O. Scanlon, "ShakeNBreak: Navigating the defect configurational landscape," *J. Open Source Software* **7**(80), 4817 (2022).
- ²⁹S. R. Kavanagh, A. G. Squires, A. Nicolson, I. Mosquera-Lois, A. M. Ganose, B. Zhu, K. Brlec, A. Walsh, and D. O. Scanlon, "Doped: Python toolkit for robust and repeatable charged defect supercell calculations," *J. Open Source Software* **9**(96), 6433 (2024).
- ³⁰L. Lang, Y.-Y. Zhang, P. Xu, S. Chen, H. J. Xiang, and X. G. Gong, "Three-step approach for computing band offsets and its application to inorganic ABX₃ halide perovskites," *Phys. Rev. B* **92**(7), 075102 (2015).
- ³¹A. M. Ganose, J. Park, A. Faghaninia, R. Woods-Robinson, K. A. Persson, and A. Jain, "Efficient calculation of carrier scattering rates from first principles," *Nat. Commun.* **12**(1), 2222 (2021).
- ³²A. Togo, L. Chaput, T. Tadano, and I. Tanaka, "Implementation strategies in phonopy and phono3py," *J. Phys.: Condens. Matter* **35**(35), 353001 (2023).
- ³³Y. Cui, K. Zhao, C. Liu, H. Xiang, H. Liang, and Y. Jia, "Band edge engineering for the improvement of open-circuit voltage: Ag-based selenized Cu₂ZnSn(SSe)₄ surface regulated by lithium," *Sol. RRL* **5**(3), 2000631 (2021).
- ³⁴D. R. Lide, editor, *CRC Handbook of Chemistry and Physics*, 81st ed. (CRC Press, Boca Raton, FL, 2000).
- ³⁵K. Zhao, H. Xiang, Y. Cui, R. Zhu, C. Liu, and Y. Jia, "Li-based selenized Cu₂ZnSnS₄ surface: Possible route to overcoming V_{oc}-deficit of kesterite solar cells," *Appl. Phys. Lett.* **118**(25), 252106 (2021).
- ³⁶S.-H. Wei, "Overcoming the doping bottleneck in semiconductors," *Comput. Mater. Sci.* **30**(3–4), 337–348 (2004).
- ³⁷C. Freysoldt, J. Neugebauer, and C. G. Van De Walle, "Electrostatic interactions between charged defects in supercells," *Phys. Status Solidi B* **248**(5), 1067–1076 (2011).
- ³⁸R. D. Shannon, "Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides," *Acta Crystallogr., Sect. A* **32**(5), 751–767 (1976).
- ³⁹H. Zhang, J. Zhang, C. Zhao, H. Xiang, K. Zhao, and C. Liu, "Exploring p-type transparent conductive materials in conventional binary compounds beyond the equilibrium doping limit," *J. Mater. Chem. C* **13**, 15471–15480 (2025).
- ⁴⁰J. Ma, S.-H. Wei, T. A. Gessert, and K. K. Chin, "Carrier density and compensation in semiconductors with multiple dopants and multiple transition energy levels: Case of Cu impurities in CdTe," *Phys. Rev. B* **83**(24), 245207 (2011).
- ⁴¹S. Chen, A. Walsh, X. Gong, and S. Wei, "Classification of lattice defects in the kesterite Cu₂ZnSnS₄ and Cu₂ZnSnSe₄ earth-abundant solar cell absorbers," *Adv. Mater.* **25**(11), 1522–1539 (2013).
- ⁴²C. Summonte, "A quasi-exact invertible equation for absorption coefficient from reflectance and transmittance measurements," *MRS Online Proc. Library* **297**, 395 (1993).
- ⁴³C. Ayutug Ava, Y. S. Ocak, O. Celik, and S. Asubay, "Deposition and characterization of Si substituted Cu₂ZnSnS₄ thin films," *Silicon* **15**(1), 451–458 (2023).
- ⁴⁴H. Kawazoe, M. Yasukawa, H. Hyodo, M. Kurita, H. Yanagi, and H. Hosono, "P-type electrical conduction in transparent thin films of CuAlO₂," *Nature* **389**(6654), 939–942 (1997).
- ⁴⁵H. Mizoguchi, M. Hirano, S. Fujitsu, T. Takeuchi, K. Ueda, and H. Hosono, "ZnRh₂O₄: A p-type semiconducting oxide with a valence band composed of a low spin state of Rh³⁺ in a 4d⁶ configuration," *Appl. Phys. Lett.* **80**(7), 1207–1209 (2002).
- ⁴⁶H. Sato, T. Minami, S. Takata, and T. Yamada, "Transparent conducting p-type NiO thin films prepared by magnetron sputtering," *Thin Solid Films* **236**(1–2), 27–31 (1993).
- ⁴⁷N. Zhang, D. Shi, X. Liu, A. Annadi, B. Tang, T. J. Huang, and H. Gong, "High performance p-type transparent LaCuOS thin film fabricated through a hydrogen-free method," *Appl. Mater. Today* **13**, 15–23 (2018).
- ⁴⁸E. L. Papadopolou, Z. A. Viskadourakis, A. V. Pennos, G. Huyberechts, and E. Aperathitis, "The effect of deposition parameters on the properties of SrCu₂O₂ films fabricated by pulsed laser deposition," *Thin Solid Films* **516**(7), 1449–1452 (2008).
- ⁴⁹D. Ahn, J. D. Song, S. S. Kang, J. Y. Lim, S. H. Yang, S. Ko, S. H. Park, S. J. Park, D. S. Kim, H. J. Chang, and J. Chang, "Intrinsically p-type cuprous iodide semiconductor for hybrid light-emitting diodes," *Sci. Rep.* **10**(1), 3995 (2020).